Aim: A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Least recently used page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

INPUT:

~

GNU nano 8.7 leastrecently.c

```c
#include<stdio.h>

int main()
{
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int frames[3];
    int i,j,k,found;
    int hit=0,fault=0;

    for(i=0;i<3;i++)
        frames[i] = -1;

    for(i=0;i<12;i++)
    {
        found = 0;

        for(j=0;j<3;j++)
        {
            if(frames[j] == pages[i])
            {
                hit++;
                found = 1;
                break;
            }
        }

        if(found == 0)
        {
            frames[i%3] = pages[i];
            fault++;
        }
    }

    printf("Page Hits = %d\n", hit);
    printf("Page Faults = %d\n", fault);

    printf("Hit Percentage = %.2f\n",(hit/12.0)*100);
    printf("Fault Percentage = %.2f\n",(fault/12.0)*100);

    return 0;
}
```

OUTPUT:

```
~

ANUJ VERMA@DESKTOP-VE8LIPH MSYS ~
$ nano leastrecently.c

ANUJ VERMA@DESKTOP-VE8LIPH MSYS ~
$ gcc leastrecently.c -o leastrecently

ANUJ VERMA@DESKTOP-VE8LIPH MSYS ~
$ ./leastrecently
Page Hits = 2
Page Faults = 10
Hit Percentage = 16.67
Fault Percentage = 83.33

ANUJ VERMA@DESKTOP-VE8LIPH MSYS ~
$ 
```